Connor Wilson

Boston, MA | wilson.conn@northeastern.edu | www.connorwilson.phd

Education

Ph.D. in Computer Science Expected June 2027
Northeastern University, Boston, MA

M.S. in Computer Science June 2024
Northeastern University, Boston, MA

B.S. in Mathematics & B.S. in Computer Science June 2022
University at Buffalo, Buffalo, NY

Research Summary

My work spans convex optimization theory and network/graph visualization to create new techniques, datasets, and tools to improve the aesthetic qualities of graph layouts.

Research, Teaching, and Professional Experience

Doctoral Researcher September 2022 – Present
Northeastern University (Advisor: Dr. Cody Dunne) Boston, MA

- Conduct and publish novel research on fast, scalable techniques for layered network visualization, improving the efficiency and usability of high-quality graph layouts.
- Maintain a documented Python codebase for generating graph visualizations using mathematical optimization models, supporting reproducible research and integration into larger data processing workflows.
- Authored 7 published papers (2 first-author) and 1 first-author paper under review as part of multiple collaborative, interdisciplinary teams.
- Taught PhD Advanced Information Visualization and Undergraduate Data Visualization courses as a teaching assistant, translating complex concepts into clear and accessible individualized support for students.

Research Engineer June 2025 – September 2025
Kostas Research Institute Boston, MA

- Created interactive, fully custom data visualizations and front-end design for global supply chain mapping application using d3.js and Svelte.js.
- Implemented network algorithms and PostgreSQL database schema for 1TB global road network dataset.

Independent Tutor January 2025 – May 2025

- Tutored undergraduate students in introductory programming, Python, and algorithms.

Research & Teaching Assistant September 2020 – May 2022
University at Buffalo (Advisor: Dr. Erdem Sariyüce) Buffalo, NY

- Implemented and improved algorithms for analyzing network substructures in a financial transaction dataset.
- Taught CSE331 Algorithms & Complexity ($4\times$).

Quality Engineering Intern June 2018 – August 2020
American Packaging Corporation Rochester, NY

- Conducted Gage Repeatability & Reproducibility studies and ANOVA for quality assurance procedures.

Publications

7. **Examining Interpretation Strategies for Multiple Forecast Visualizations with Two and Four Forecasts**
   Padilla L., Fygenson R., ***Wilson, C.***, Potter K., and Castro S. *ACM Conference on Human Factors in Computing Systems—CHI*. 2026.

6. **GLaDOS: Graph Layout algorithm Datasets for Open Science**
   Di Bartolomeo S., ***Wilson, C.***, Puerta E., Crnovrsanin T., Frings A., and Dunne C. *Journal of Visualization and Interaction—JoVI*. 2025. journalovi.org/2024-dibartolomeo-benchmark/

5. **Fast and Readable Layered Network Visualizations Using Large Neighborhood Search**
   Wilson, C., Crnovrsanin T., Puerta E., and Dunne C. *IEEE Transactions on Visualization and Computer Graphics*. 2025. doi.org/10.1109/TVCG.2025.3537898.

4. **Evaluating and extending speedup techniques for optimal crossing minimization in layered graph drawings**
   Wilson, C., Puerta E., Di Bartolomeo S., Crnovrsanin T., and Dunne C. *IEEE Transactions on Visualization and Computer Graphics—VIS/TVCG*. 2024. 10.1109/TVCG.2024.3456349.

3. **Evaluating Graph Layout Algorithms: A Systematic Review of Methods and Best Practices**
   Di Bartolomeo S., Crnovrsanin T., Saffo D., Puerta E., ***Wilson, C.***, and Dunne C. *Computer Graphics Forum—EuroVis/CGF*. 2023. doi.org/10.1111/cgf.15073.

2. **A collection of benchmark datasets for evaluating graph layout algorithms**
   Di Bartolomeo S., Puerta E., ***Wilson, C.***, Crnovrsanin T., and Dunne C. *Poster at Graph Drawing and Network Visualization—GD Posters*. 2023. **Best Poster Honorable Mention*

1. **Indy Survey Tool: A framework to unearth correlations in survey data**
   Crnovrsanin T., Di Bartolomeo S., ***Wilson, C.***, and Dunne C. *Proc. IEEE Visualization Conference—VIS*. 2023. doi.org/10.1109/VIS54172.2023.00038.

Under Submission

- **CINDER: An expressive framework for optimal layered graph layouts**
  Wilson, C., Puerta E., and Dunne C. *Under submission to IEEE Transactions on Visualization and Computer Graphics*.

Honors

Best Poster Honorable Mention 2023
International Symposium on Graph Drawing and Network Visualization, Palermo, Italy

Program Valedictorian 2022
Department of Mathematics, University at Buffalo, Buffalo, NY

Professional Service

Conference Student Volunteer Chair 2026
IEEE VIS 2026, Boston, MA

PhD Admissions Committee Student Representative 2025 – 2026
Khoury College of Computer Science, Northeastern University, Boston, MA

Conference Student Volunteer 2024 – 2025
IEEE VIS 2024, Tampa, FL; IEEE VIS 2025, Vienna, Austria

Conference/Journal Reviewer 2024 – 2025
IEEE PacificVis; SoftwareX

PhD Admissions Committee Volunteer 2023
Khoury College of Computer Science, Northeastern University, Boston, MA

Lab Operations Manager 2022 – 2025
Data Visualization Lab, Northeastern University, Boston, MA

Technical Skills & Interests

Programming Languages: Python | JavaScript | C | HTML | CSS
Technologies and Database Systems: Gurobi | React | Svelte | d3.js | Jupyter | SQL | MongoDB | PgAdmin
Interests: Running | Chess Puzzles | Electronic Music | Coffee | Buffalo Sports Fan | Mentoring